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Hypothesis scorecard: verdicts as the reports state them

Three hypotheses were frozen before any model was built, each with a falsification condition. One of them fired. Every verdict below is a statement about a model, not a vehicle.

HYPOTHESIS, VERBATIM	VERDICT	THE ONE NUMBER	WHAT LIMITS IT
H0 Minimising peak heat flux alone does not necessarily produce the thermally safest re-entry solution.	Supported in the tested domain	-38.3% / +114.1 K peak flux / bondline between the two ends of a 41-point entry-angle sweep; 818 counterexample pairs M1 M1-20260902T130547Z	Falsifiable and once falsified: at a 40 mm insulator the bondline rose 8.8 K over a whole entry and the two metrics decoupled (NR-03). Fidelity 0, one parameter, so the $\rho = -1$ rank correlation is tautological; the monotonicity is the evidence. Restated at Fidelity 1 and under uncertainty by H1's numbers.
H1 Joint optimisation of peak heat flux and in-depth TPS response can identify safer feasible designs than peak-heat-only optimisation.	Supported, narrowly (see §10.4)	-14.9 K knee vs peak-flux-only optimum at the bondline, for +21.5 kW/m²; paired on shared draws -14.31 K (s.d. 1.45), below zero in 3000 of 3000 draws M4-OPT-20260920T205252Z · M7-UQ-20260920T233048Z	The front trades the objectives through entry angle only; diameter, bluntness and cone angle are pinned by a mass fence (the knee's shield is 346 of 350 kg), the CFD-hull edge and a box bound. It is the same mechanism as H0, not evidence that joint optimisation finds a better capsule shape.
H2 Surrogate-assisted and AI-guided adaptive-fidelity search can approach the Pareto frontier with fewer expensive CFD evaluations than unguided or brute-force search.	NOT SUPPORTED	0 of 5 seeds reached the pre-declared target (95% of reference hypervolume = 0.3264); the rule needed 3; no arm reached it M6 M6-AF-20260920T211148Z	The target was beyond the study's own search budget: all arm-seeds pooled reach 94.1% of the reference (NR-35), and extra CFD moved the no-CFD arm's score by 0.0001. The AI-guided adaptive arm was declared and not run, so the 'AI-guided' clause is untested. Not contradicted in general; not supported here.
M5 Not a hypothesis: the pre-declared AI-vs-conventional comparison. LLM engineering agent against the best of 4 conventional optimisers, 5 seeds, rule declared first: gain ≥ 0.005 in hypervolume AND $\alpha = 0.05$.	MIXED	+0.0283 +0.0085 +0.0036 at 50 evaluations AI helped at 100 evaluations AI helped at 200 evaluations no measured difference M5 M5-ABL-20260920T211134Z	At 200 evaluations every agent seed beat every comparator seed, and the verdict is still 'no measured difference' because the mean gain fell under the effect size declared in advance. n = 5, one model, one easy problem; the early lead cannot be separated from prior knowledge, and the AI-plus-adaptive-fidelity arm has no result.

Hypothesis text and verdict wording: reports/final/AETHER_paper.md §4 (parsed). H0 numbers: results/M1/M1-20260902T130547Z/candidates.csv and the M1 report. H1: results/M4/M4-OPT-20260920T205252Z/summary.json, results/M7/M7-UQ-20260920T233048Z/paired_difference.json.

H2: results/M6/M6-AF-20260920T211148Z/summary.json (h2, criteria_declared) and reports/milestones/M6_adaptive_fidelity.md (pooled reach). M5: results/M5/M5-ABL-20260920T211134Z/summary.json (criteria). Caveats paraphrase the paper's §4, §10.4, §11.2, §12 and §17.

Within tested assumptions. Nothing here is established about a real vehicle or material; the coupon experiment has not been run.